UNIVERSITY OF THE PEOPLE

BIOL 1301-01 - AY2026-T5

Submitted by: Mubashir Ali

Instructor: Modupe Stella Ayilara

Date: Aug 18, 2026

Ecology, Energy Flow, and Human Impacts on Biodiversity

Part 1: Ecology of a Natural Environment

A. Ecological Organization in Deosai National Park, Pakistan

For this assignment, I selected Deosai National Park in Gilgit-Baltistan, Pakistan. Deosai, also known as the "Land of Giants," is one of the world's highest alpine plateaus. It consists of alpine grasslands, wetlands, streams, lakes, and snow-covered mountains. The park is famous for its rich biodiversity, including the Himalayan brown bear, snow leopard, golden marmot, ibex, wolves, foxes, numerous migratory birds, and a wide variety of alpine plants. This ecosystem plays an important role in conserving endangered wildlife and maintaining ecological balance.



Figure 1. Deosai National Park, Gilgit-Baltistan, Pakistan. The park contains alpine grasslands, wetlands, lakes, and diverse wildlife that form a unique mountain ecosystem.

The different levels of ecological organization in Deosai include:

Individual (Organism): An individual organism is a single living thing. For example, one Himalayan brown bear represents an individual organism.

Population: A population consists of individuals of the same species living in the same area. All Himalayan brown bears living in Deosai National Park form one population.

Community: A community includes all populations of different species living and interacting in the same area. In Deosai, the community consists of brown bears, marmots, ibex, wolves, birds, insects, grasses, flowers, and microorganisms.

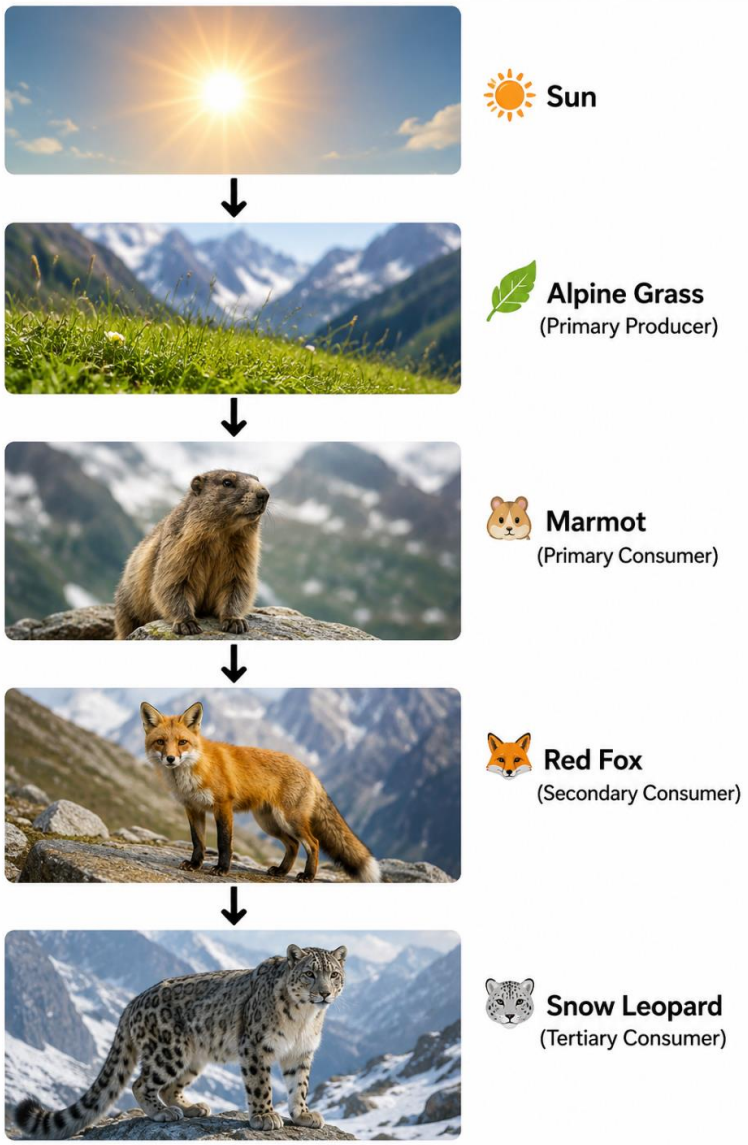
Ecosystem: An ecosystem includes both living (biotic) and non-living (abiotic) components interacting together. In Deosai, the ecosystem includes plants, animals, soil, rocks, rivers, air, sunlight, water, and climatic conditions that support life (Clark et al., 2020).



Figure 2. Examples of organisms and biodiversity found in Deosai National Park.

B. Food Chain and Energy Flow

A simple food chain in Deosai National Park is:



Energy enters the ecosystem from the sun. Alpine grasses capture solar energy through photosynthesis and act as the primary producers. Marmots feed on grasses and serve as primary consumers (herbivores). Red foxes prey on marmots and act as secondary consumers, while snow leopards occupy the highest trophic level as tertiary consumers or apex predators.

Energy flows in one direction through the food chain. Only about ten percent of the energy available at one trophic level is transferred to the next level, while the remaining energy is lost as heat during metabolic processes. Therefore, producers contain the greatest amount of energy, whereas apex predators receive the least amount (Clark et al., 2020). This explains why ecosystems support many producers but relatively few top predators.

C. Ecological Interaction: Predation

One important ecological interaction in Deosai is predation between the snow leopard and Himalayan ibex or marmots.



Snow leopards hunt herbivores to obtain energy. Predation regulates prey populations and prevents overgrazing of alpine vegetation. Healthy predator populations help maintain ecosystem stability by preventing any one species from becoming excessively abundant. If predators disappeared, herbivore populations would increase rapidly, leading to excessive consumption of plants, habitat degradation, and reduced biodiversity. Therefore, predation contributes significantly to ecological balance and species diversity.

Part 2: Human Activities Affecting Biodiversity

A. Impact of Human Activities

Two important human activities affecting biodiversity in Pakistan are deforestation and climate change.

The first activity is deforestation. Forests are cleared for agriculture, urban expansion, fuelwood, and infrastructure development. Habitat destruction forces wildlife to migrate or perish, reducing biodiversity and disrupting ecological interactions. Deforestation also increases soil erosion, decreases carbon storage, alters water cycles, and contributes to climate change.

The second activity is climate change, largely driven by greenhouse gas emissions. Pakistan has experienced rising temperatures, glacier melting, unpredictable rainfall, floods, droughts, and shifting habitats. These environmental changes threaten mountain ecosystems such as Deosai National Park by altering vegetation patterns and reducing suitable habitats for many species.

Climate change also affects migration, breeding, and food availability for wildlife (Clark et al., 2020).

Learning ecological and biological principles has improved my understanding of these issues. Ecology demonstrates that organisms are interconnected through food webs, nutrient cycles, and energy flow. Damage to one component of an ecosystem affects many other organisms. Understanding trophic levels, biodiversity, and ecological relationships highlights why protecting habitats is essential for maintaining ecosystem health.

B. Personal Reflection

I believe biodiversity conservation should be a national priority because healthy ecosystems support human life as well as wildlife. Deforestation and uncontrolled greenhouse gas emissions should not continue at their current rate. Instead, sustainable forestry, habitat restoration, renewable energy, and stronger conservation policies should be expanded.

Pakistan has already established protected areas such as Deosai National Park, which demonstrates the importance of conservation. However, additional public awareness, environmental education, stricter enforcement of environmental laws, and community participation are necessary to protect biodiversity for future generations.

This unit helped me understand that every organism has an ecological role. Producers capture energy, consumers transfer energy through food chains, predators regulate populations, and decomposers recycle nutrients. Human activities can either preserve or damage these natural processes. As future scientists and responsible citizens, we should promote sustainable development that balances economic growth with environmental protection.

References

Clark, M. A., Choi, J., & Douglas, M. (2020). *Biology 2e*. OpenStax. https://openstax.org/details/books/biology-2e

Khan Academy. (2021, January 27). *Human impacts on ecosystems | Biodiversity and human impacts* [Video]. YouTube. https://www.youtube.com/

MooMooMath and Science. (2019, May 2). *Ecological relationships—Competition, predator and prey, symbiosis* [Video]. YouTube. https://www.youtube.com/